Experiment-8 : (linked stack and linked queue)

Q1) Write a C program that uses functions to implement linked stack on single linked list.

```
#include<stdio.h>

#include<stdlib.h>

struct node{

   int data;

   struct node *next;

};

struct node *top=NULL;

void push(){

   struct node *temp;

   int x;

   temp=(struct node*)malloc(sizeof(struct node));

   printf("Enter value: ");

   scanf("%d",&x);

   temp->data=x;

   temp->next=top;

   top=temp;

}

void pop(){

   struct node *temp;

   if(top==NULL){

      printf("Stack Underflow\n");

      return;

   }

   temp=top;

   top=top->next;
```

```
    free(temp);
}
void display(){
    struct node *temp=top;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}
int main(){
    int ch;
    while(1){
        printf("1.Push 2.Pop 3.Display 4.Exit\n");
        scanf("%d",&ch);
        switch(ch){
            case 1: push(); break;
            case 2: pop(); break;
            case 3: display(); break;
            case 4: exit(0);
        }
    }
}
```

Q2) Write a C program that uses functions to implement linked queue on single linked list.

```
#include<stdio.h>

#include<stdlib.h>

struct node{

   int data;

   struct node *next;

};

struct node *front=NULL,*rear=NULL;

void enqueue(){

   struct node *temp;

   int x;

   temp=(struct node*)malloc(sizeof(struct node));

   printf("Enter value: ");

   scanf("%d",&x);

   temp->data=x;

   temp->next=NULL;

   if(rear==NULL){

      front=rear=temp;

      return;

   }

   rear->next=temp;

   rear=temp;

}

void dequeue(){

   struct node *temp;

   if(front==NULL){

      printf("Queue Underflow\n");
```

```
        return;
    }
    temp=front;
    front=front->next;
    if(front==NULL)
        rear=NULL;
    free(temp);
}
void display(){
    struct node *temp=front;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}
int main(){
    int ch;
    while(1){
        printf("1.Enqueue 2.Dequeue 3.Display 4.Exit\n");
        scanf("%d",&ch);
        switch(ch){
            case 1: enqueue(); break;
            case 2: dequeue(); break;
            case 3: display(); break;
            case 4: exit(0);
        }
    }  }
```